

Fraud Detection Pipeline

Summary Report | March 2026

1. Overview

End-to-end fraud detection pipeline: data quality checks, 39-feature engineering (multi-window velocity, cyclical encoding, outlier capping), 4 classifiers (Logistic Regression, Random Forest, XGBoost, LightGBM), stacking ensemble, SMOTE comparison, overfitting mitigation, SHAP explainability, isotonic calibration, and recall optimisation at 30-60%. Best model: XGBoost (PR-AUC=0.150, ROC-AUC=0.944).

2. Data Quality & Preparation

	Raw	After Dedup	Columns	Fraud Rate
Train	157,306	113,905	48 (44 after label drop)	0.11%
Test	110,998	74,792	44	Unknown

Pipeline: null normalisation (sentinel strings -> NaN) -> deduplication (fraud-priority) -> encoding fixes -> type casting -> identifier removal -> quality report. Temporal hold-out: test starts Apr 2020, train from Feb 2020.

3. Feature Engineering (31 numeric + 8 categorical)

Category	Features	Rationale
Multi-window velocity	card_txn_{1,3,7,14,30}d	Burst patterns at diff. scales
Recency	hours_since_last_txn	Rapid-fire fraud detection
Card stats	card_avg_amount, amount_deviation	Anomalous amounts
Card-merchant	card_merchant_first, card_unique_merchants	New merchant risk
Geographic	card_billing_mismatch, geoip_billing_mismatch	Cross-border fraud
Cyclical temporal	hour_sin/cos, dow_sin/cos + raw	Smooth time-of-day signal
Amount	log_euramount, euramount_capped	Skew + outlier control
Domain	domain_freq (email domain frequency)	Synthetic identities
Categorical (8)	cardbrand, cardtype, decline_type, ...	One-hot encoded

Leakage-safe: lookup tables built from training fold only. Rare categories binned (freq < 0.5%); outliers capped at 1st/99th percentile. Temporal 80/20 split: Train=91,124, Val=22,781 (0.16% fraud, 36 fraud cases).

4. Class Imbalance Handling

Strategies: class_weight='balanced' (LR, RF), scale_pos_weight (XGBoost), is_unbalance=True (LightGBM). SMOTE experiment: resampled 91,124 -> 181,986 rows (fraud 131 -> 90,993). Result: class weights (PR-AUC=0.150) outperformed SMOTE (PR-AUC=0.043) — SMOTE over-smooths minority-class boundaries in extreme-imbalance regimes.

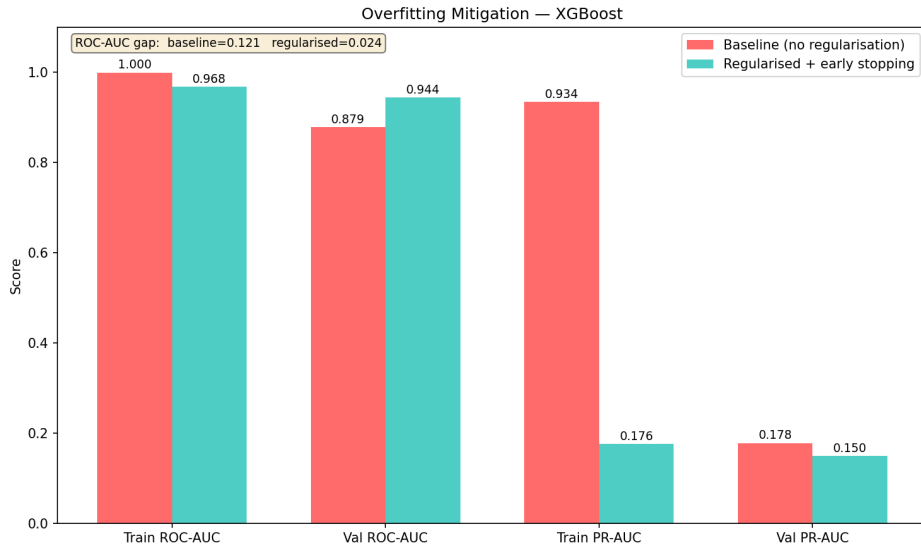
5. Model Results

Model	ROC-AUC	PR-AUC	F1	Val Recall	Train/Val ROC gap
LogisticRegression	0.936	0.028	0.018	89%	0.019
RandomForest	0.943	0.075	0.038	56%	0.038
XGBoost*	0.944	0.150	0.017	92%	0.024
LightGBM	0.608	0.006	0.016	44%	0.348
Stacking	---	0.077	---	---	---

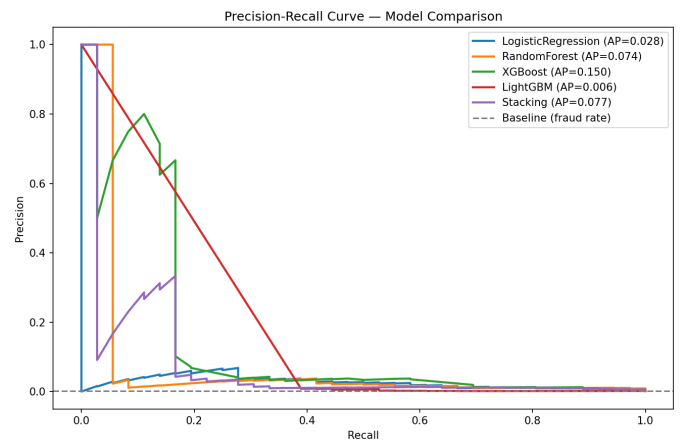
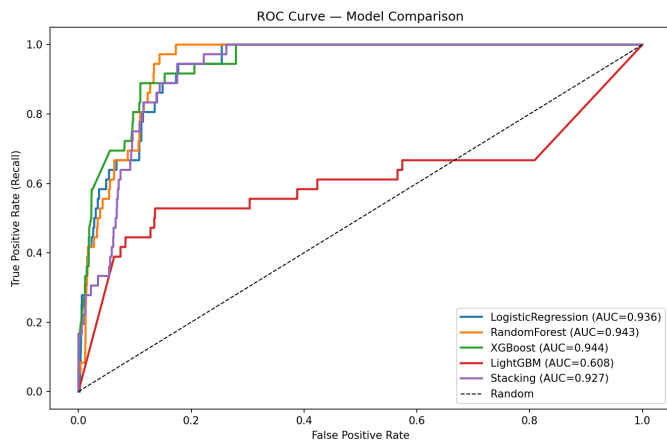
* Best model selected by PR-AUC. Stacking includes only models with ROC-AUC >= 0.60.

6. Overfitting Detection & Mitigation

Baseline XGBoost (no regularisation, max_depth=12, 500 trees): Train ROC=0.9998, Val ROC=0.879, gap=0.121. Regularised XGBoost (max_depth=6, subsample=0.8, colsample=0.8, early stopping): Train ROC=0.968, Val ROC=0.944, gap=0.024. Regularisation reduces the overfit gap by 5x while improving validation performance.



7. Model Comparison



XGBoost dominates on PR-AUC (0.150), the most informative metric for extreme imbalance. High ROC-AUC across LR/RF/XGB (0.93-0.94) masks the difficulty visible in PR-AUC. LightGBM underperforms (ROC-AUC=0.61) despite safety-net retraining.

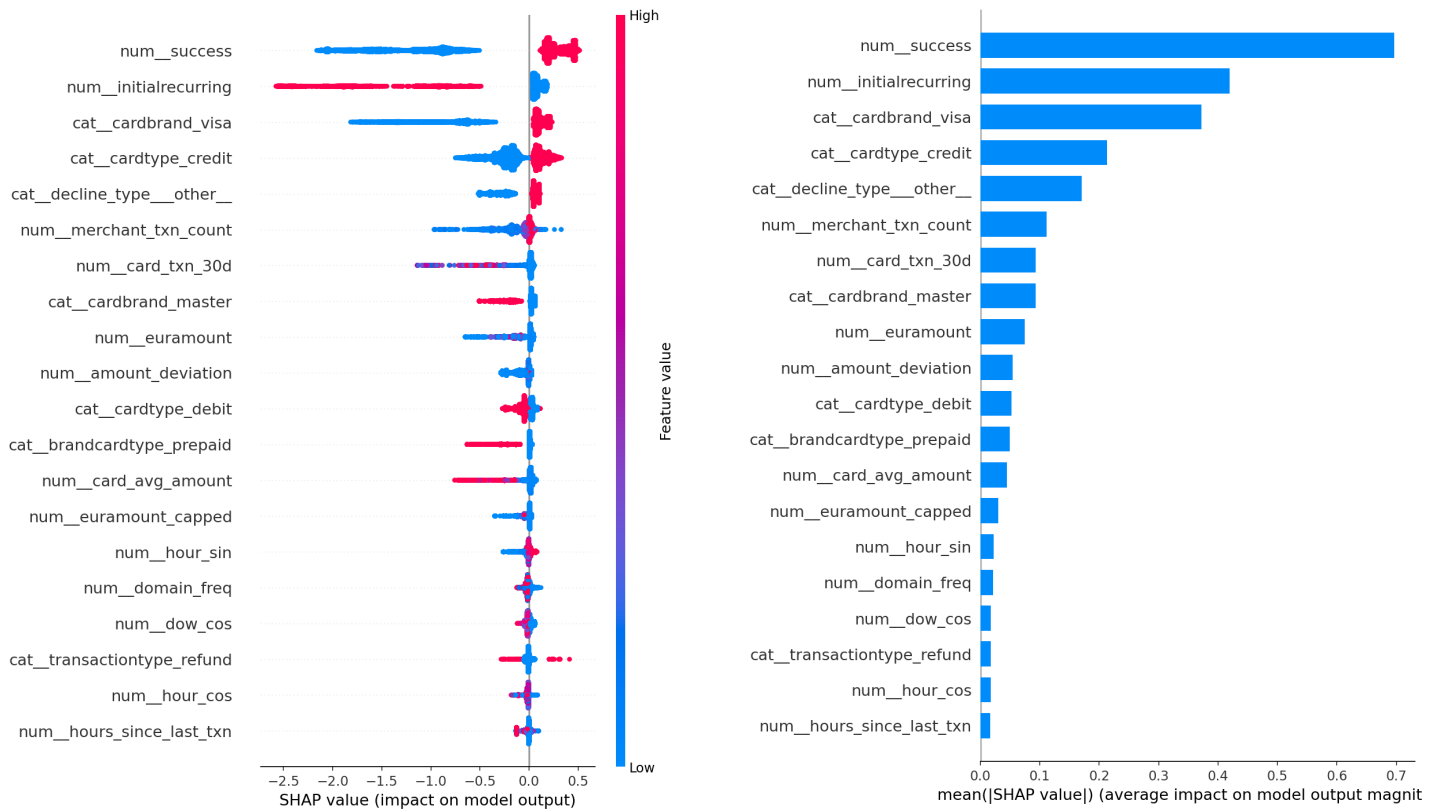
8. Recall Optimisation (30%-60%)

XGBoost (Best Model) — Recall Operating Points:

Target	Threshold	Actual Recall	Precision	F1
30%	0.7887	33.3%	0.0426	0.0755
35%	0.7645	47.2%	0.0376	0.0697
40%	0.7645	47.2%	0.0376	0.0697
45%	0.7645	47.2%	0.0376	0.0697
50%	0.7621	58.3%	0.0376	0.0706
55%	0.7621	58.3%	0.0376	0.0706
60%	0.7537	69.4%	0.0192	0.0375

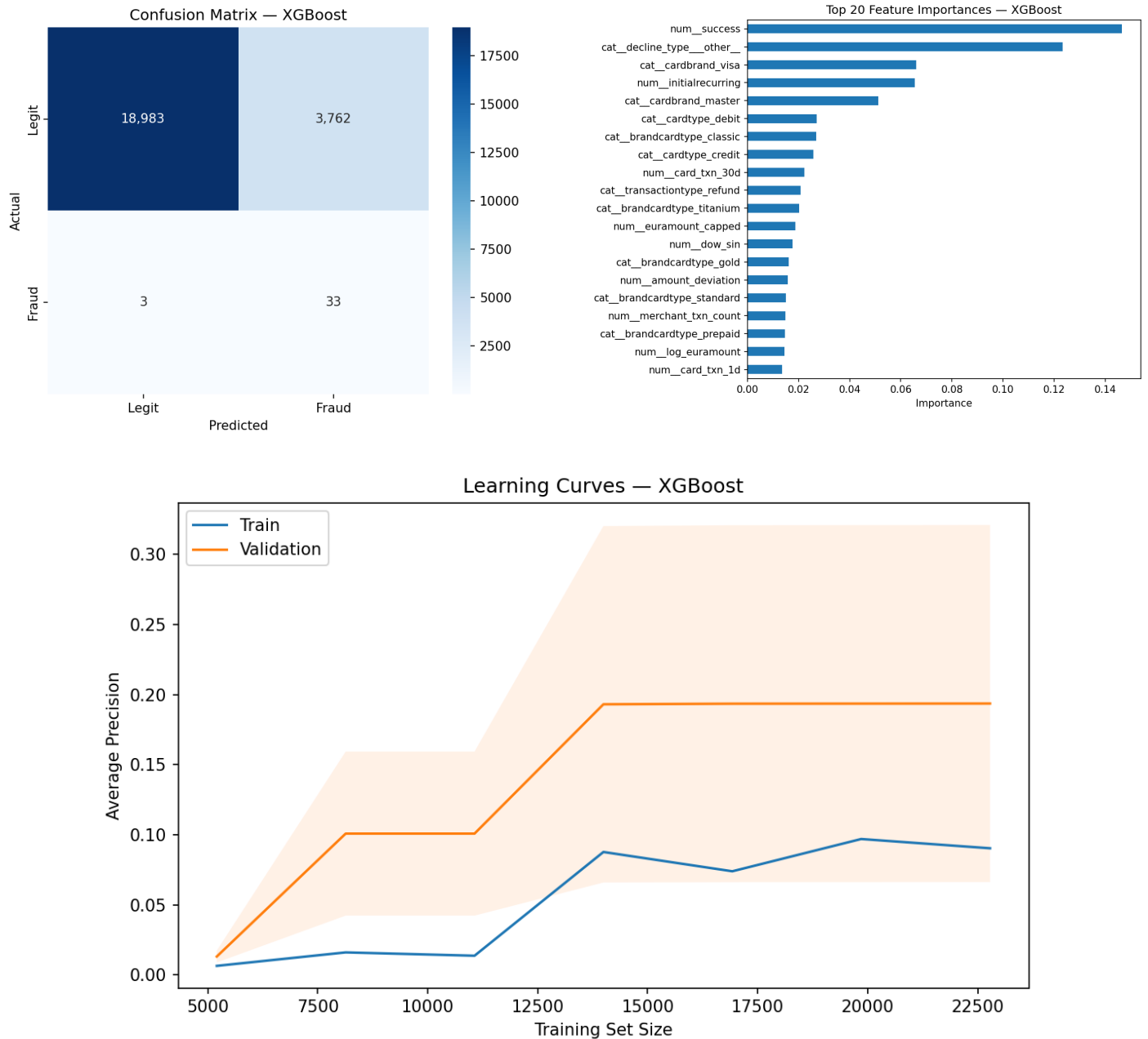
Test predictions: threshold=0.027 (targeting ~50% recall), flagging 2,341/74,792 transactions (3.13%) as potentially fraudulent.

9. Explainability (SHAP)



SHAP TreeExplainer on the best XGBoost model. Beeswarm plot (left) shows per-feature impact direction; bar plot (right) shows mean absolute SHAP importance. Top features provide transparent, auditable explanations for each fraud prediction.

10. Best Model Detail (XGBoost)



11. Conclusions

- XGBoost selected as best model (PR-AUC=0.150, ROC-AUC=0.944) with isotonic calibration applied
- Class weights outperform SMOTE (PR-AUC 0.150 vs 0.043) for this extreme-imbalance regime
- Regularisation reduces overfit gap from 0.121 to 0.024 (5x improvement)
- SHAP explainability provides transparent, auditable per-prediction reasoning
- Stacking ensemble (PR-AUC=0.077) underperforms standalone XGBoost due to weaker base learners
- Serialised artifacts (best_model.joblib, lookup_tables.joblib) enable deployment without retraining